

WASP-140b

R-2306

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-140_201012 · created 2026-08-17T23:30:05+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459134.92 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.014
Transit depth (Rp/Rs)^2	2.88 +/- 0.49 [%]
Orbital Inclination (inc)	84.17 +/- 0.4
Airmass coefficient 1 (a1)	1.349 +/- 0.012
Airmass coefficient 2 (a2)	-0.172 +/- 0.036
Scatter in the residuals of the lightcurve fit is	0.86 %
Best Comparison Star	None
Optimal Aperture	5.46
Optimal Annulus	13.26
Transit Duration (day)	0.0762 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.17 ± 0.014 vs 0.1432 (deviation 1.79σ)
Duration fit vs published	0.0762 d vs 0.0604 d (deviation 2.96σ)
Mid-time O–C	-0.6 min
Depth SNR	11.4
Residual scatter	0.86 %

Light curve

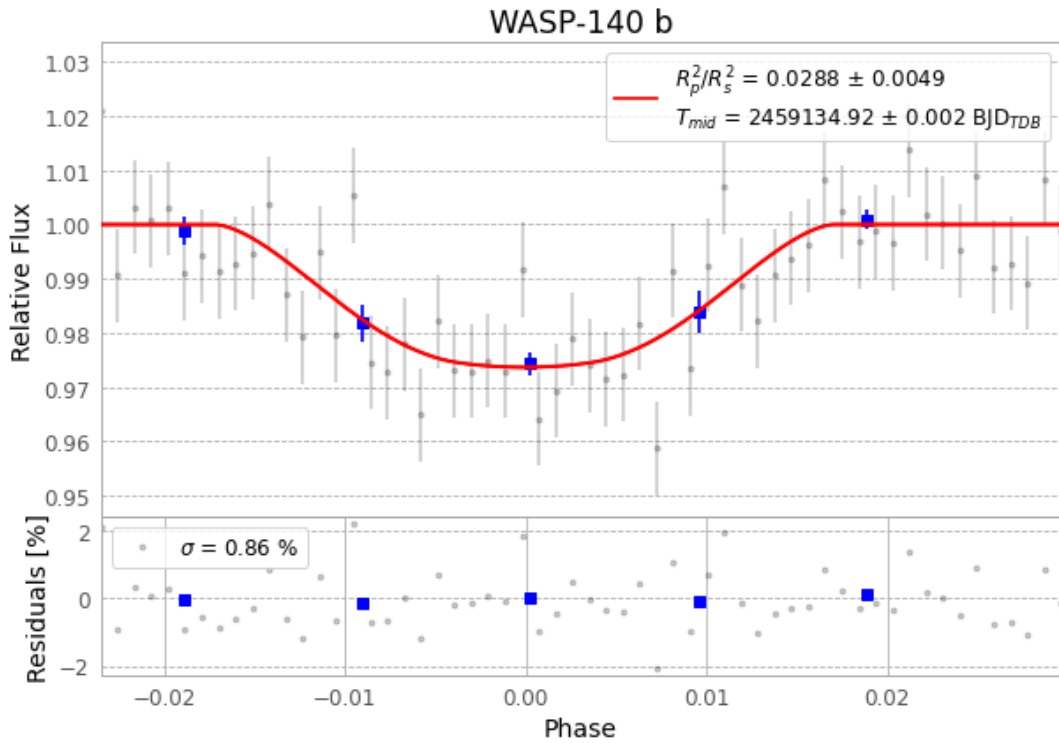


Figure 1 — FinalLightCurve_WASP-140 b_2020-10-12.png (SHA-256 24440e5b25106e96...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [92, 241], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c919524c40a21acb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

58 FITS frames (38 MB), WASP-140201012084215.FITS → WASP-140201012113312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-140-201012.log (35eec40f2873...), FinalLightCurve_WASP-140 b_2020-10-12.png (24440e5b2510...), FinalLightCurve_WASP-140 b_2020-10-12.csv (bed2af0a02a4...)

Compute resources

Wall time 184.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 23:30Z.